

# INTEGRATED ELECTRICAL SYSTEMS OF SOLAR PHOTOVOLTAIC TECHNOLOGY

LABORATORY 1

GROUP 2



**TÉCNICO** LISBOA

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Electrical Configurations of Photovoltaic Cell Arrays

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2025/2026 – 2nd Semester, P3

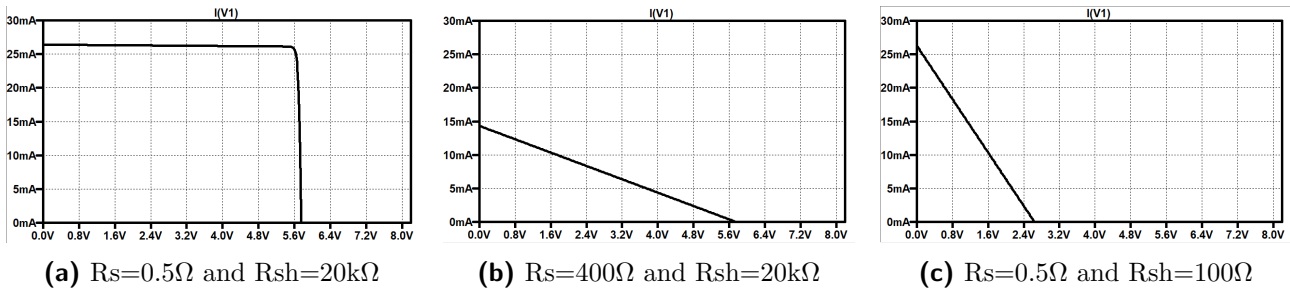
## 1 Introduction

The main goal of this laboratory is to test different configurations of photovoltaic cells. Each one has its advantages and disadvantages and may be useful in different circumstances. With access to a solar source and 4 photovoltaic cells, the configurations tested were: Single Cell, Series, Parallel, Series-Parallel and TCT. Data was acquired using an oscilloscope and the I(V) curve was constructed in MATLAB. For each configuration, an I-V characteristic curve, a power-voltage (P-V) plot, and a fill factor (FF) visualization were generated.

Additionally, a single diode five parameter (1D5P) equivalent model was implemented in MATLAB to extract the photovoltaic parameters from the experimental I-V curves and compare the model results with the measured behaviour of the different configurations.

Before proceeding with the laboratory data acquisition, a preliminary simulation was performed in *LTSpice* to establish the expected behaviour. A 1D5P equivalent circuit was designed using ideal parameters (low series resistance,  $R_s$ , and high shunt resistance,  $R_{sh}$ ) to serve as a baseline, as shown in Figure 1a. To better visualize the impact of the internal losses, the value of  $R_s$  was increased and the value of  $R_{sh}$  was decreased, as shown in Figure 1b and Figure 1c, respectively.

As expected, the simulation verified that an increase of  $R_s$  causes a reduction in the slope of the I-V curve near the  $V_{oc}$  region, affecting the  $I_{sc}$  value. In the other hand, a decrease in  $R_{sh}$  affects the  $V_{oc}$  value.



**Figure 1:** I-V curve impact of  $R_s$  and  $R_{sh}$

## 2 Results

All tests were conducted using a projector as the light source, placed at 39 cm from the table and the measured irradiance was  $1015 \text{ W/m}^2$ .

The setup consisted of the solar cell connected in series to a variable resistor and a  $1 \Omega$  resistor to measure current. The I(V) curve was captured performing a load sweep using the variable resistor. As the resistor was manually adjusted, the operating point of the solar cell transitioned through the entire curve from the short circuit point (lowest resistance) to the open circuit point (highest resistance). Using this method, it is not possible to go exactly to the  $I_{sc}$  and  $V_{oc}$  because the lowest resistance available is different than a resistance of  $0 \Omega$  and the highest resistance is different than  $\infty \Omega$ .

This module should not be called cell as a solar cell is the individual silicon wafer that converts light into electricity. Because the module is composed of many cells, the appropriate name is module. As this is the lowest configuration on the laboratory, the module is often referred as a cell.

Due to lack of time, different modules' characteristics weren't tested but different modules have different characteristics.

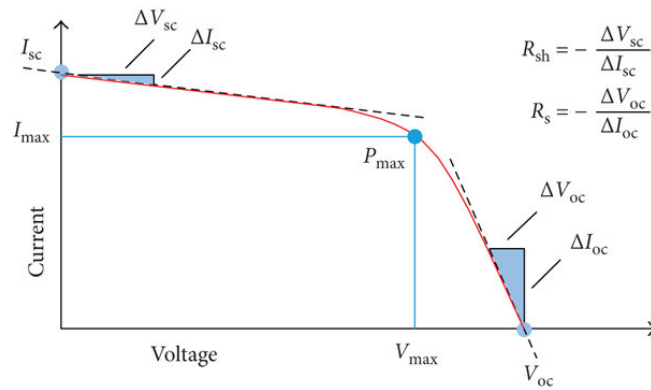
In order to perform the I(V) curve, MATLAB was used to fit a smooth trend line into the points of the oscilloscope. A figure with Power vs Voltage was also performed in order to know the maximum

power point of operation, where the output power is maximum. With the values of the maximum power point, the fill factor can be computed using this formula:

$$FF (\%) = \frac{P_{\max}}{V_{oc} \cdot I_{sc}} \times 100$$

$$FF (\%) = \frac{V_{\max} \cdot I_{\max}}{V_{oc} \cdot I_{sc}} \times 100$$

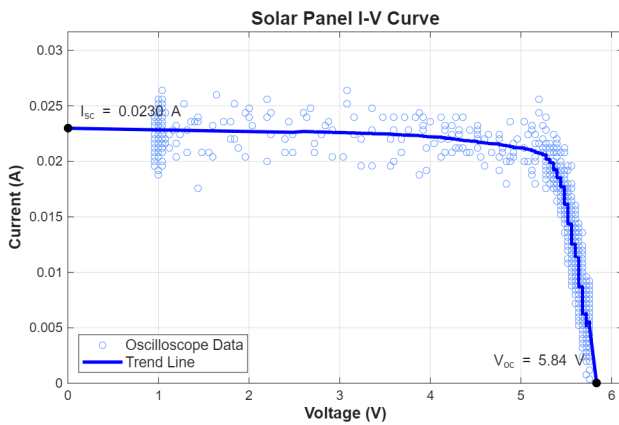
The values of the Shunt Resistor and Series Resistor were also determined by the negative of the slope of the I-V curve.



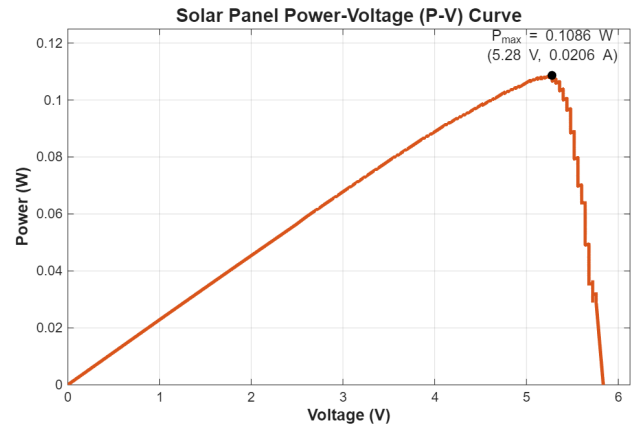
**Figure 2:**  $R_{sh}$  and  $R_s$  visualization

## 2.1 I(V) and P(V) Photovoltaic Module Characteristics

For the single module configuration the I(V) curve was the following. All values are presented below.



**Figure 3:** Single module I-V curve

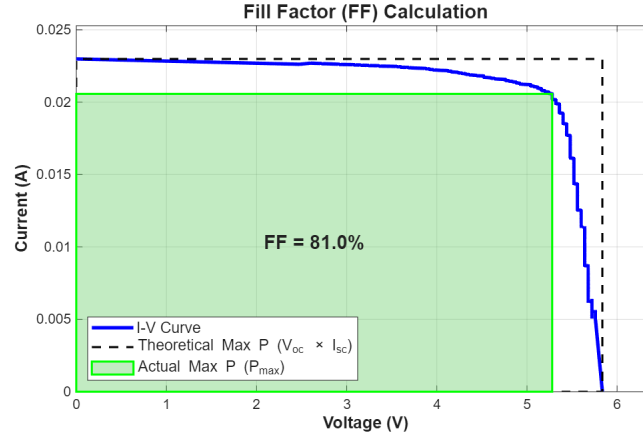


**Figure 4:** Single module Power-Voltage curve

The data sheet of the solar panel indicates that the typical peak power is 550 mW, the voltage at peak power is 5.5 V and current at peak power is 100 mA. This is very different compared with the results obtained of a peak power of 109 mW with 5.28 V and 20 mA. This is explained by the difficulty on recreating the Standard Test Conditions (STC) that the manufacturer uses to test the solar panels. In STC the irradiance should be  $1000 \text{ W/m}^2$  and the cell temperature should be  $25^\circ\text{C}$ . Tests were

conducted with an irradiance level of  $1015 \text{ W/m}^2$  but temperature was not controlled precisely, with values going to  $49^\circ\text{C}$  sometimes. Also the  $R_s$  is  $20 \Omega$  contributing to power losses.

Next the fill factor was computed using the formulas presented before.



**Figure 5:** Single module Fill Factor visualization

A fill factor of 81% is a good value as it means the solar panel is operating near its theoretical max power point.

**Table 1:** Extracted electrical parameters of the configuration

| Solar Panel Parameter              | Value            |
|------------------------------------|------------------|
| Open-Circuit Voltage ( $V_{oc}$ )  | 5.8367 V         |
| Short-Circuit Current ( $I_{sc}$ ) | 0.0230 A         |
| Maximum Power ( $P_{max}$ )        | 0.1086 W         |
| Voltage at $P_{max}$ ( $V_{mp}$ )  | 5.2800 V         |
| Current at $P_{max}$ ( $I_{mp}$ )  | 0.0206 A         |
| Fill Factor (FF)                   | 80.97 %          |
| Shunt Resistance ( $R_{sh}$ )      | 6744.97 $\Omega$ |
| Series Resistance ( $R_s$ )        | 20.39 $\Omega$   |

Efficiency can also be computed using this formula:

$$\eta = \frac{P_{max}}{P_{in}} \times 100\%$$

Using the obtained values:

- $P_{max} = 0,1086 \text{ W}$
- $G = 1015 \text{ W/m}^2$
- $A = 0,070 \times 0,055 = 0,00385 \text{ m}^2$

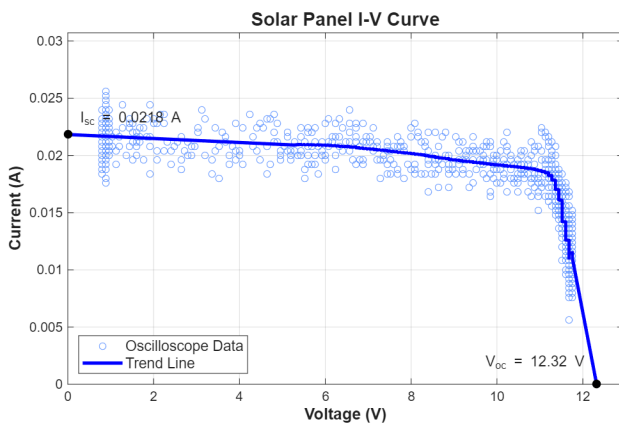
$$\eta = \frac{0,1086}{1015 \cdot 0,00385} \times 100\% \approx 2,78\%$$

This value of efficiency is low compared to the 17% announced by the manufacturer. This can be justified by the spectral mismatch of the artificial light source, which fails to replicate the AM1.5 solar spectrum; a tilting angle that reduces efficiency; higher cell temperature; and high series resistance.

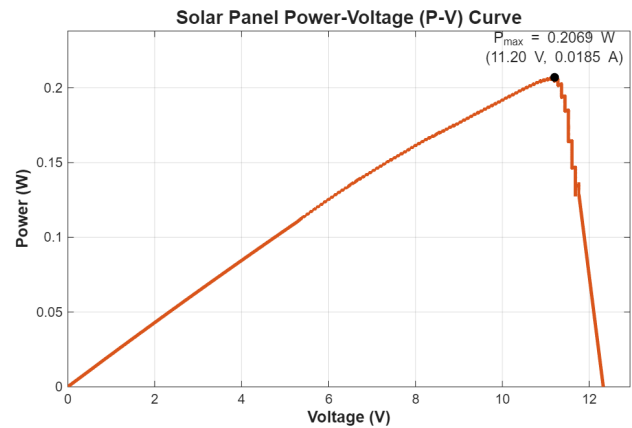
## 2.2 I(V) and P(V) Characteristics of 2 Photovoltaic Modules Connected in Series

For the 2 modules in series configuration the I(V) curve was the following. All values are presented bellow.

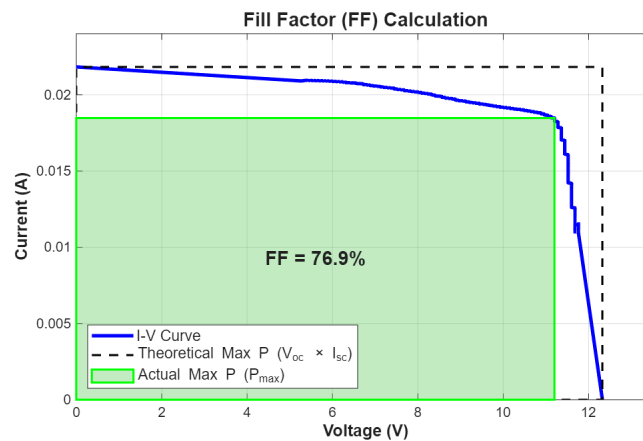
When 2 modules are connected in series, current flows through both units, making the total differential potential equal to the sum of the independent voltages. Meanwhile the current remains the same. This is demonstrated with the value of  $V_{oc}$  that is approximately twice the  $V_{oc}$  when just one module is connected.



**Figure 6:** I-V curve of two solar modules connected in series



**Figure 7:** Power-Voltage (P-V) curve of the same series connection



**Figure 8:** In-series Fill Factor visualization

**Table 2:** Extracted electrical parameters of the configuration

| Solar Panel Parameter              | Value            |
|------------------------------------|------------------|
| Open-Circuit Voltage ( $V_{oc}$ )  | 12.3219 V        |
| Short-Circuit Current ( $I_{sc}$ ) | 0.0218 A         |
| Maximum Power ( $P_{max}$ )        | 0.2069 W         |
| Voltage at $P_{max}$ ( $V_{mp}$ )  | 11.2000 V        |
| Current at $P_{max}$ ( $I_{mp}$ )  | 0.0185 A         |
| Fill Factor (FF)                   | 76.94 %          |
| Shunt Resistance ( $R_{sh}$ )      | 5728.43 $\Omega$ |
| Series Resistance ( $R_s$ )        | 54.33 $\Omega$   |

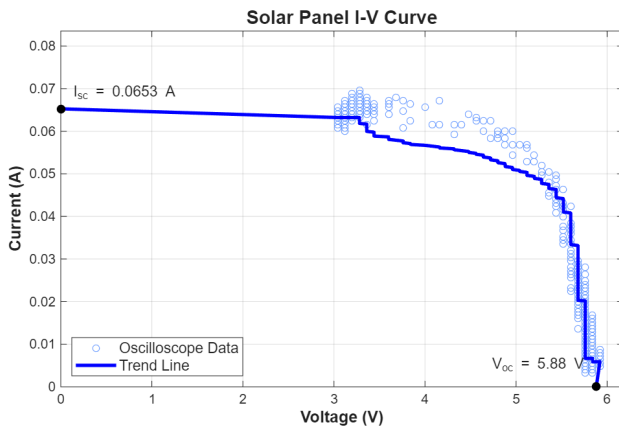
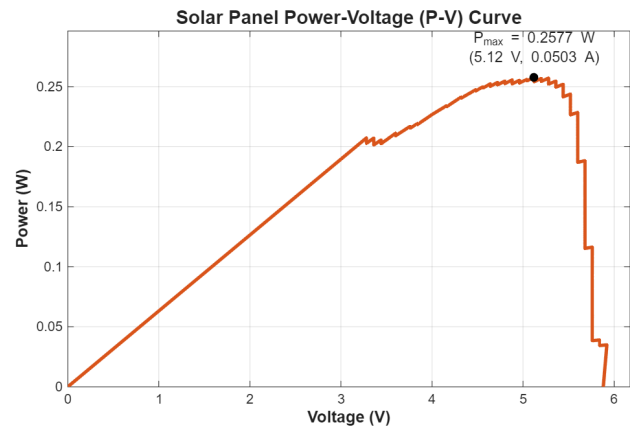
$$\eta = \frac{0,2069}{1015 \cdot 2 \times 0,00385} \times 100\% \approx 2,65\%$$

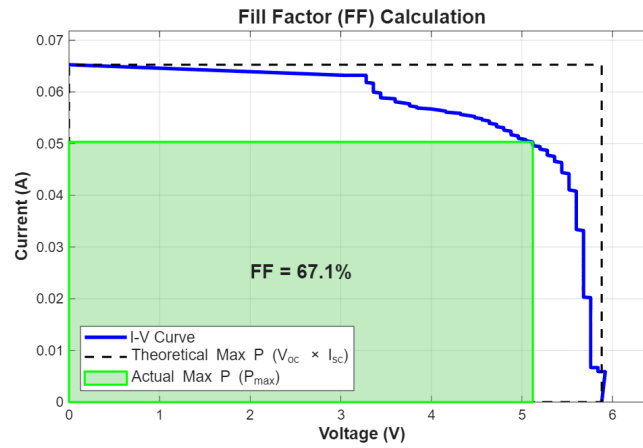
### 2.3 I(V) and P(V) Characteristics of 2 Photovoltaic Modules Connected in Parallel

When 2 modules are connected in parallel, the positive terminals are connected together as well as the negative ones. Because of this, the voltage of the configuration is the same as the voltage of one single model. Meanwhile the total current is the sum of the currents produced by each module. This is demonstrated with the value of  $I_{sc}$  that is approximately twice the  $I_{sc}$  when just one module is connected.

For the 2 modules in parallel configuration, the I(V) curve was the following. All values are presented below.

The experimental data for this configuration contains some errors, for example the lowest value of voltage is around 3V compared with around 1V on other experiments. The lack of data points in the low-voltage region introduced significant uncertainty into the curve-fitting process.

**Figure 9:** I-V curve of two solar modules connected in parallel**Figure 10:** Power-Voltage (P-V) curve of the same parallel connection



**Figure 11:** Parallel Fill Factor visualization

**Table 3:** Extracted electrical parameters of the configuration

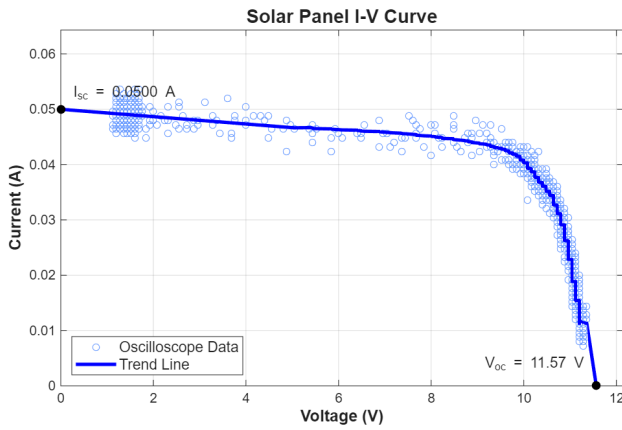
| Solar Panel Parameter              | Value                  |
|------------------------------------|------------------------|
| Open-Circuit Voltage ( $V_{oc}$ )  | 5.8807 V               |
| Short-Circuit Current ( $I_{sc}$ ) | 0.0653 A               |
| Maximum Power ( $P_{max}$ )        | 0.2577 W               |
| Voltage at $P_{max}$ ( $V_{mp}$ )  | 5.1200 V               |
| Current at $P_{max}$ ( $I_{mp}$ )  | 0.0503 A               |
| Fill Factor (FF)                   | 67.15 %                |
| Shunt Resistance ( $R_{sh}$ )      | Infinite (Forced Flat) |
| Series Resistance ( $R_s$ )        | 8.36 $\Omega$          |

$$\eta = \frac{0,2577}{1015 \cdot 2 \times 0,00385} \times 100\% \approx 3.30\%$$

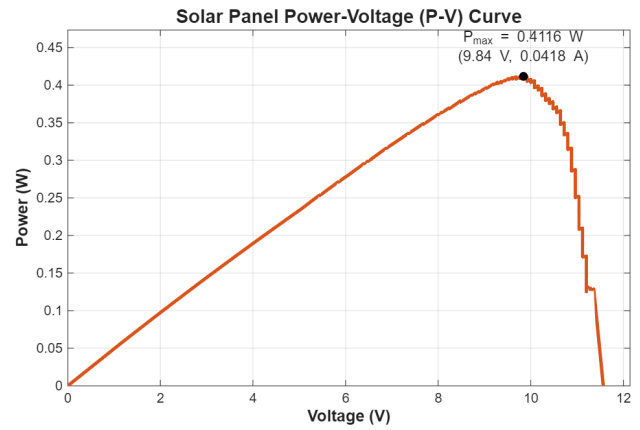
## 2.4 I(V) and P(V) Characteristics of a Series-Parallel 2x2 Group of Photovoltaic Modules

For the 4 modules in series-parallel configuration, the I(V) curve was the following. All values are presented bellow.

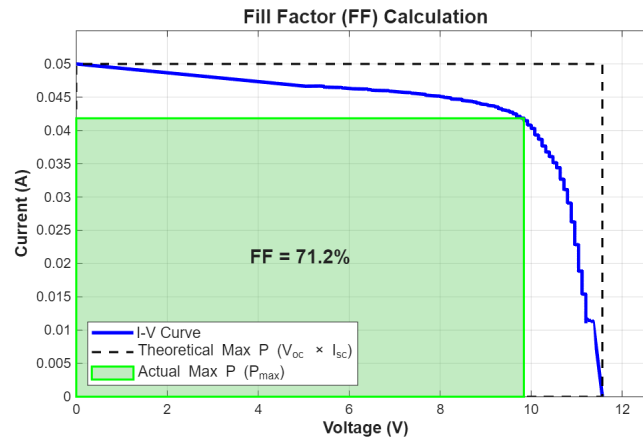
When 4 modules are connected in series-parallel, the total current and voltage correspond to the sum of currents and voltages. This is demonstrated with the values of  $I_{sc}$  and  $V_{oc}$  that are approximately twice the  $I_{sc}$  and  $V_{oc}$  when just one module is connected.



**Figure 12:** I-V curve of solar modules connected in series-parallel



**Figure 13:** Power-Voltage (P-V) curve of the same series-parallel connection



**Figure 14:** Series-Parallel Fill Factor visualization

**Table 4:** Extracted electrical parameters of the configuration

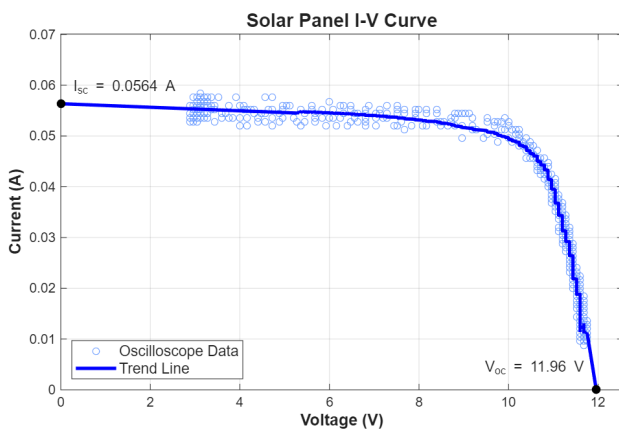
| Solar Panel Parameter              | Value            |
|------------------------------------|------------------|
| Open-Circuit Voltage ( $V_{oc}$ )  | 11.5662 V        |
| Short-Circuit Current ( $I_{sc}$ ) | 0.0500 A         |
| Maximum Power ( $P_{max}$ )        | 0.4116 W         |
| Voltage at $P_{max}$ ( $V_{mp}$ )  | 9.8400 V         |
| Current at $P_{max}$ ( $I_{mp}$ )  | 0.0418 A         |
| Fill Factor (FF)                   | 71.18 %          |
| Shunt Resistance ( $R_{sh}$ )      | 1504.95 $\Omega$ |
| Series Resistance ( $R_s$ )        | 27.36 $\Omega$   |

$$\eta = \frac{0,4116}{1015 \cdot 4 \times 0,00385} \times 100\% \approx 2,63\%$$

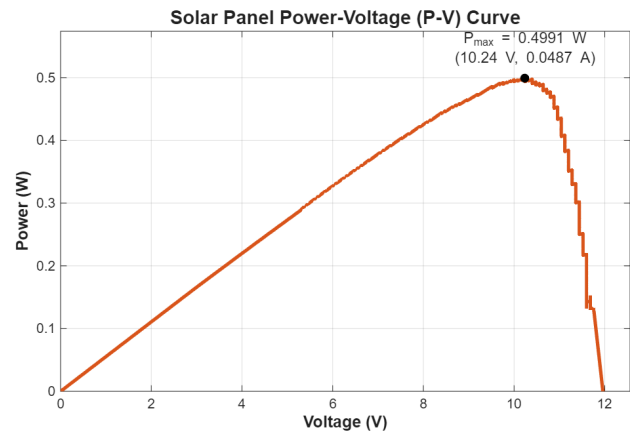
## 2.5 I(V) and P(V) Characteristics of a TCT Configuration with 4 Photovoltaic Modules

When 4 modules are connected in TCT, voltage and current are the same as the 4 modules in series-parallel. In series-parallel configurations the cells are organized in series and then the different arrays of in-series cells are connected in parallel with other arrays of the same size. TCT and series-parallel connections are the same except TCT adds a connection between all junctions, providing a higher level of redundancy. If a module is shaded, the current can redistribute through the parallel paths in that same row, effectively bypassing the shaded area.

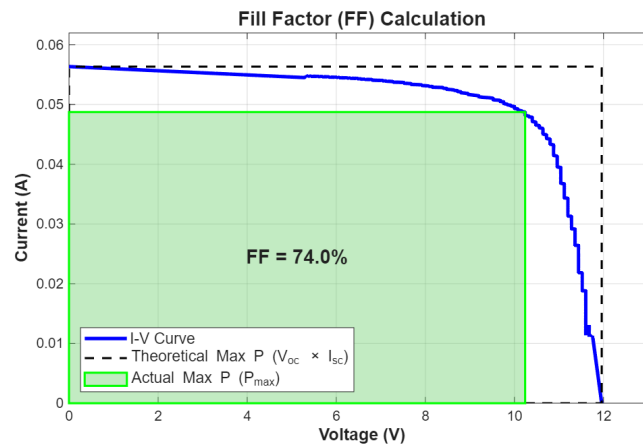
For the 4 modules in TCT configuration, the I(V) curve was the following. All values are presented below.



**Figure 15:** I-V curve of the solar modules in TCT configuration



**Figure 16:** Power-Voltage (P-V) curve of the solar modules in TCT configuration



**Figure 17:** TCT Fill Factor visualization

**Table 5:** Extracted electrical parameters of the configuration

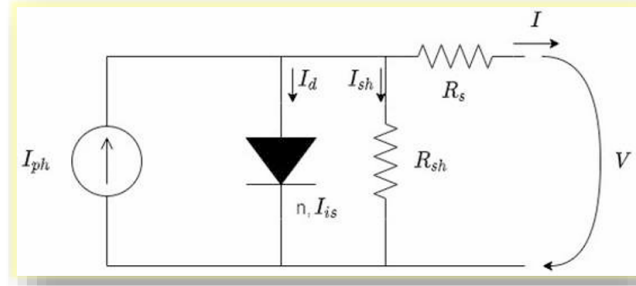
| Solar Panel Parameter              | Value            |
|------------------------------------|------------------|
| Open-Circuit Voltage ( $V_{oc}$ )  | 11.9595 V        |
| Short-Circuit Current ( $I_{sc}$ ) | 0.0564 A         |
| Maximum Power ( $P_{max}$ )        | 0.4991 W         |
| Voltage at $P_{max}$ ( $V_{mp}$ )  | 10.2400 V        |
| Current at $P_{max}$ ( $I_{mp}$ )  | 0.0487 A         |
| Fill Factor (FF)                   | 74.05 %          |
| Shunt Resistance ( $R_{sh}$ )      | 2869.33 $\Omega$ |
| Series Resistance ( $R_s$ )        | 23.46 $\Omega$   |

$$\eta = \frac{0,4991}{1015 \cdot 4 \times 0,00385} \times 100\% \approx 3,19\%$$

It's possible to see that this configuration had the highest efficiency and it was capable of producing around 90 mW of power more than the series-parallel configuration. This proves that the TCT configuration is more redundant and provides the best power output from all the 5 setups tested.

## 2.6 Photovoltaic model

In order to model the electrical behaviour of the photovoltaic modules, a single-diode five-parameter (1D5P) equivalent model was used. This model represents the photovoltaic cell as a current source in parallel with a diode and a shunt resistance, together with a series resistance accounting for internal losses as shown in 18.

**Figure 18:** 1D5P Equivalent Model

The I–V characteristic of the photovoltaic model can be described by:

$$I = I_{ph} - I_0 \left( e^{\frac{V - IR_s}{nV_t}} - 1 \right) - \frac{V + IR_s}{R_{sh}} \quad (1)$$

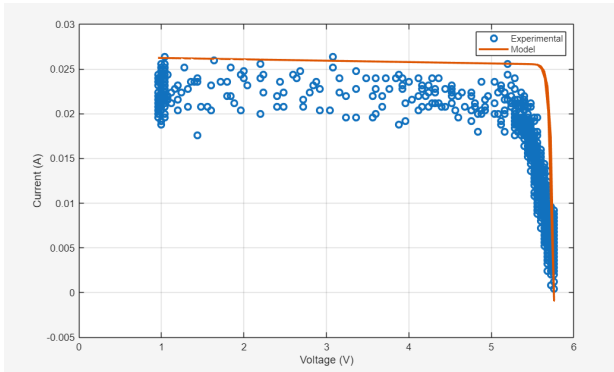
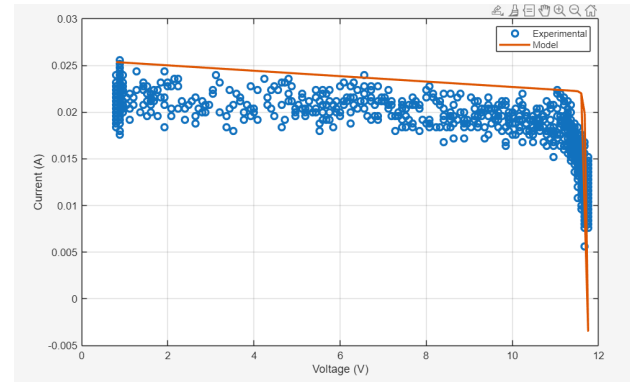
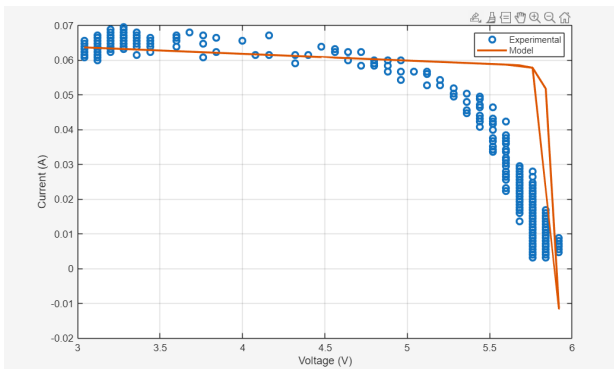
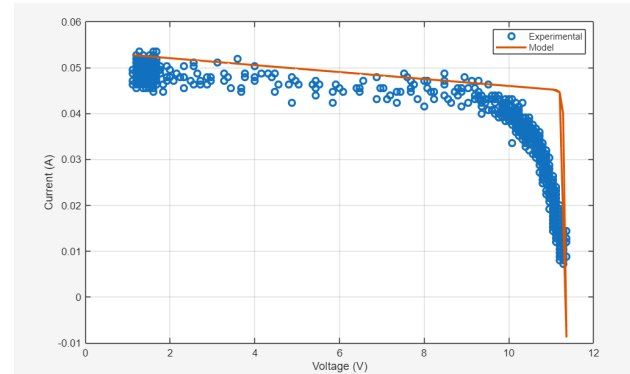
where  $I_{ph}$  is the photocurrent,  $I_0$  is the diode saturation current,  $R_s$  is the series resistance,  $R_{sh}$  is the shunt resistance,  $n$  is the diode ideality factor and  $V_t$  is the thermal voltage.

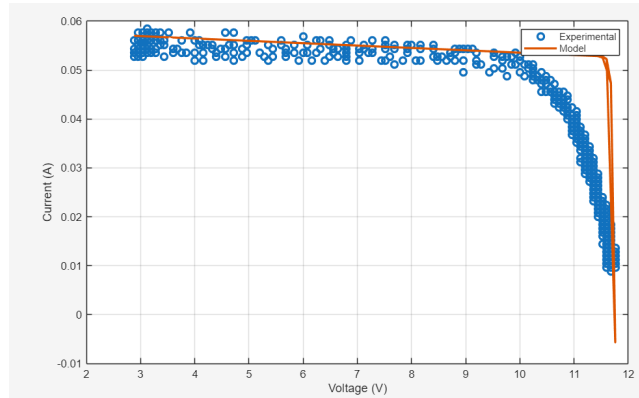
The parameters of the model were extracted from the experimental I–V curves obtained in the previous sections using MATLAB. The fitting quality was evaluated using the Root Mean Square Error (RMSE) between the experimental data and the model prediction.

Table 6 presents the obtained parameters for the different photovoltaic configurations analysed in this work.

**Table 6:** Extracted parameters of the photovoltaic equivalent model for the different configurations.

| Configuration         | $I_{ph}$ (A) | $I_0$ (A)               | $R_s$ ( $\Omega$ ) | $R_{sh}$ ( $\Omega$ ) | $n$ | RMSE                  |
|-----------------------|--------------|-------------------------|--------------------|-----------------------|-----|-----------------------|
| Single module         | 0.0264       | $1.33 \times 10^{-76}$  | 21.88              | 6500                  | 1.3 | $1.14 \times 10^{-2}$ |
| Series                | 0.0256       | $5.26 \times 10^{-154}$ | 32.14              | 3450                  | 1.3 | $9.91 \times 10^{-3}$ |
| Parallel              | 0.0696       | $3.04 \times 10^{-78}$  | 16.00              | 516.67                | 1.3 | $3.69 \times 10^{-2}$ |
| Series-Parallel (2x2) | 0.0536       | $1.59 \times 10^{-148}$ | 39.66              | 1322.22               | 1.3 | $2.22 \times 10^{-2}$ |
| TCT configuration     | 0.0584       | $1.20 \times 10^{-153}$ | 34.33              | 2066.67               | 1.3 | $2.97 \times 10^{-2}$ |

**Figure 19:** Experimental and modelled I-V characteristic for a single photovoltaic module.**Figure 20:** Experimental and modelled I-V characteristic for two modules connected in series.**Figure 21:** Experimental and modelled I-V characteristic for two modules connected in parallel.**Figure 22:** Experimental and modelled I-V characteristic for the 2x2 series-parallel configuration.



**Figure 23:** Experimental and modelled I-V characteristic for the TCT configuration.

The comparison between the experimental I-V curves and the curves obtained from the photovoltaic model shows that the general behaviour of the modules is reproduced for all analysed configurations. This agreement can be observed visually in the figures and is also reflected in the RMSE values presented in Table 6.

Among the analysed configurations, the series connection presents the lowest RMSE ( $9.91 \times 10^{-3}$ ), indicating a closer correspondence between the model and the experimental data. The single module configuration also shows a relatively small error ( $1.14 \times 10^{-2}$ ). In contrast, the parallel configuration presents the highest RMSE ( $3.69 \times 10^{-2}$ ). In the corresponding I-V plot, the experimental points appear grouped in horizontal bands, which is likely related to the measurement resolution and to the higher currents present in this configuration.

The series-parallel and TCT configurations show intermediate RMSE values ( $2.22 \times 10^{-2}$  and  $2.97 \times 10^{-2}$ ), and the fitted curves follow the overall trend of the experimental points, although with slightly larger deviations than in the series case. These results indicate that the selected equivalent model captures the main electrical behaviour of the photovoltaic modules for the different electrical configurations analysed.

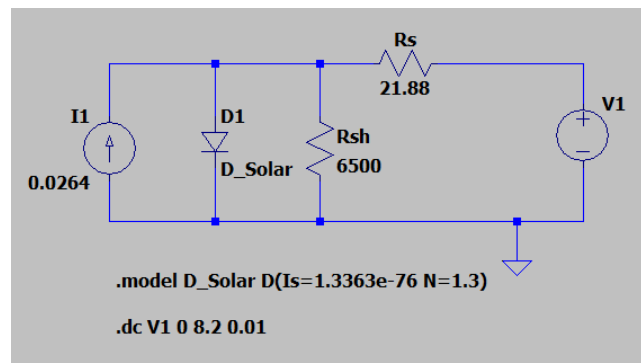
From the obtained parameters it can be observed that configurations with modules connected in parallel present higher photocurrent values, since the currents from each module add together. On the other hand, series connections mainly increase the voltage while keeping the current approximately constant.

The series-parallel ( $2 \times 2$ ) and TCT configurations present intermediate behaviours, combining both effects. Overall, the selected single-diode equivalent model provides an adequate representation of the photovoltaic behaviour of the modules for the different electrical configurations analysed.

### 3 Extraordinary Questions

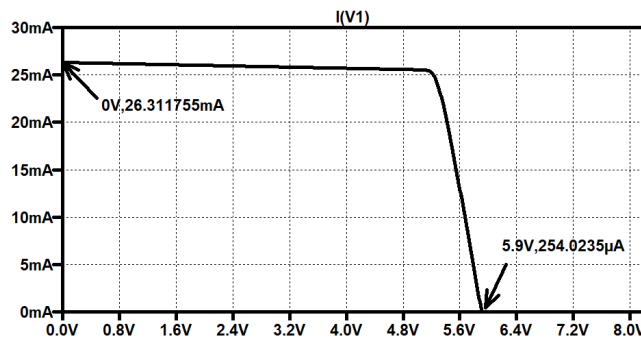
In order to verify the experimental measurements and the theoretical model output for the single module configuration, a simulation of the 1D5P equivalent circuit was performed in LTSpice, as shown in Figure 24.

The circuit was parametrized using the values previously extracted in Section 5.6.



**Figure 24:** 1D5P equivalent circuit in LTSpice

By plotting the simulated I-V curve, it is possible to verify that the behaviour aligns with both the experimental and theoretical results, since the simulated short-circuit current,  $I_{sc}$ , is 26.312 mA and the open-circuit voltage,  $V_{oc}$ , is 5.9V.



**Figure 25:** LTSpice simulation using the extracted parameters from the 1D5P model

**Table 7:** Comparison of results between Experimental data and LTSpice simulation.

| Parameter     | Experimental | LTSpice | Relative Error (%) |
|---------------|--------------|---------|--------------------|
| $I_{SC}$ (mA) | 23.0         | 26.31   | 14.39              |
| $V_{OC}$ (V)  | 5.8367       | 5.9     | 1.08               |